

Worksheet 17 – Understanding Rand Index and Adjusted Rand Index

(a) Using the Iris dataset as an example, explain what this sentence means: The Rand Index is used to compare clustering results against an external criteria (ground truth).

(b) Suppose the results of two clustering methods involving 6 observations:

Method 1 (ground-truth) : {1, 1, 2, 2, 3, 3} [this means first two observations is grouped together in cluster 1, next two observations in cluster 2 and the last two in cluster 3]

Method 2: {1, 1, 1, 2, 2, 2}

Use R's EMCluster package and the RRand() function to calculate the rand index and adjusted rand index. Write you code and results here:

(c) Validate R's answer for Rand Index in part (b) by calculating Rand Index by hand.

For your reference, from Wikipedia:

Given a set of n elements $S = \{o_1, \dots, o_n\}$ and two partitions of S to compare, $X = \{X_1, \dots, X_r\}$, a partition of S into r subsets, and $Y = \{Y_1, \dots, Y_s\}$, a partition of S into s subsets, define the following:

- a , the number of pairs of elements in S that are in the **same** subset in X and in the **same** subset in Y
- b , the number of pairs of elements in S that are in **different** subsets in X and in **different** subsets in Y
- c , the number of pairs of elements in S that are in the **same** subset in X and in **different** subsets in Y
- d , the number of pairs of elements in S that are in **different** subsets in X and in the **same** subset in Y

The Rand index, R , is:^{[1][2]}

$$R = \frac{a + b}{a + b + c + d} = \frac{a + b}{\binom{n}{2}}$$

(d) By looking at the formula of the Rand Index, can you deduce the largest possible value for Rand Index? Under what conditions will the Rand Index have the largest possible value? What is the smallest possible value for the Rand Index?

(e) Suppose you randomly assign the 6 observations to 3 clusters (each cluster has at least one observation). When compared against the ground-truth cluster result of {1, 1, 2, 2, 3, 3}, do you expect the Rand Index to be 0? Use R to test it out. Write your code here:

Note: It will be a nice property if Rand Index has a value close to 0 when observations are grouped purely by chance – but that is not the case. The Adjusted Rand Index is created to rectify this deficiency of the Rand Index.

(e) To adjust the Rand Index for chance (to address the issue in part (e)), we use the Adjusted Rand Index. Repeat what you did in part (e) using R but calculate the Adjusted Rand Index instead. Is the Adjusted Rand Index close to 0 now? [Note that the Adjusted Rand Index can be negative in value].

Because the Adjusted Rand Index is the adjusted-for-chance version of the Rand Index, its use is preferred over the Rand Index. This probably explains why your textbook mentioned the Adjusted Rand Index but did not mention the Rand Index at all. Calculating the Adjusted Rand Index is tedious and you will not be tested on it for this course. However, you are expected to be able to perform Rand Index calculations. Just to keep this worksheet complete, let's try to calculate Adjusted Rand Index by hand for the data in part (b). From Wikipedia:

The contingency table [\[edit \]](#)

Given a set S of n elements, and two groupings or partitions (e.g. clusterings) of these elements, namely $X = \{X_1, X_2, \dots, X_r\}$ and $Y = \{Y_1, Y_2, \dots, Y_s\}$, the overlap between X and Y can be summarized in a contingency table $[n_{ij}]$ where each entry n_{ij} denotes the number of objects in common between X_i and Y_j : $n_{ij} = |X_i \cap Y_j|$.

$X \backslash Y$	Y_1	Y_2	$\dots$	Y_s	sums
X_1	n_{11}	n_{12}	$\dots$	n_{1s}	a_1
X_2	n_{21}	n_{22}	$\dots$	n_{2s}	a_2
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
X_r	n_{r1}	n_{r2}	$\dots$	n_{rs}	a_r
sums	b_1	b_2	$\dots$	b_s	

Definition [\[edit \]](#)

The original Adjusted Rand Index using the Permutation Model is

$$ARI = \frac{\sum_{ij} \binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}}{\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}}$$

where n_{ij} , a_i , b_j are values from the contingency table.